Fermeate

revolutionizing precision fermentation from within

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Problem

The CapEx of expanding capacity is massive

Bottleneck

pre-fermentation

during fermentation

optimization happens here

cannot control

Demand

A need for more capacity is URGENT

the CapEx of expanding capacity is too constraining and makes keeping up with demand unsustainable

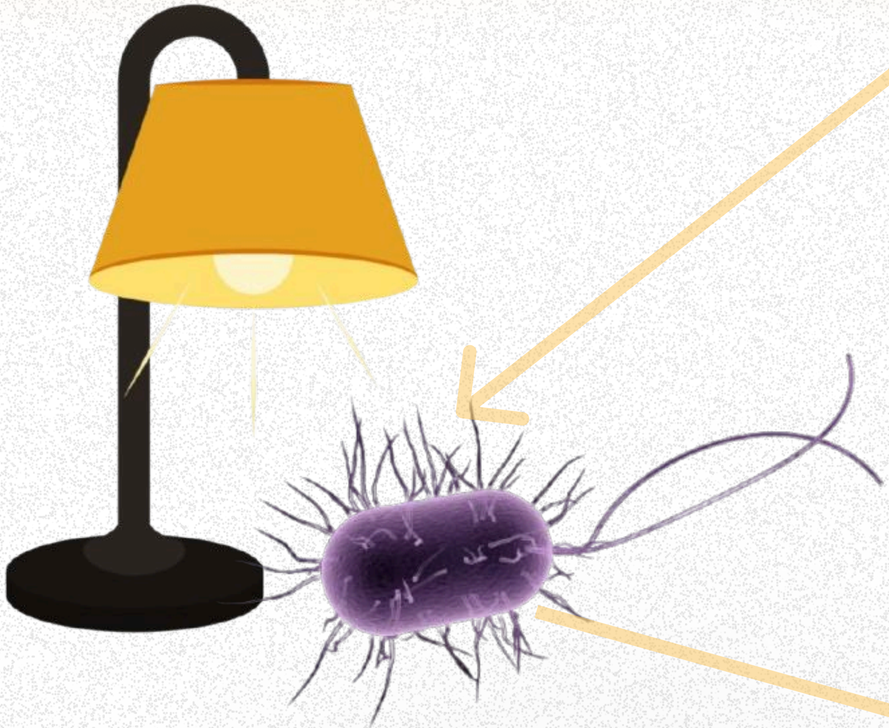
1000x

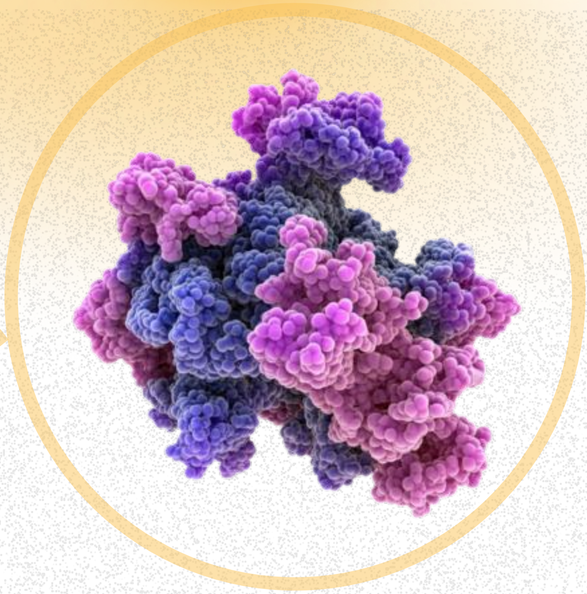
more capacity needed by 2030

200%

exceed in capacity demanded

Technology Breakthrough





gene expressed

gene suppressed

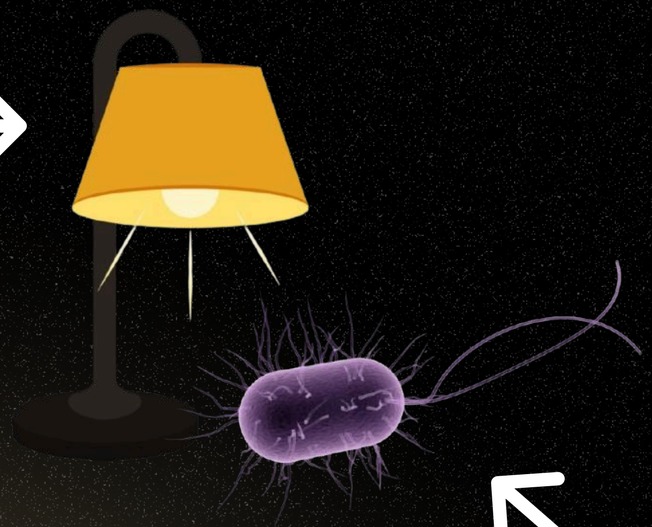
- light is tunable and reversible
- continuous, real-time control of gene expression
- enables microbes to operate at optimal efficiency

Founder

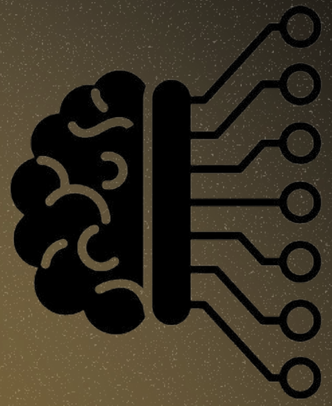
Founding Researcher

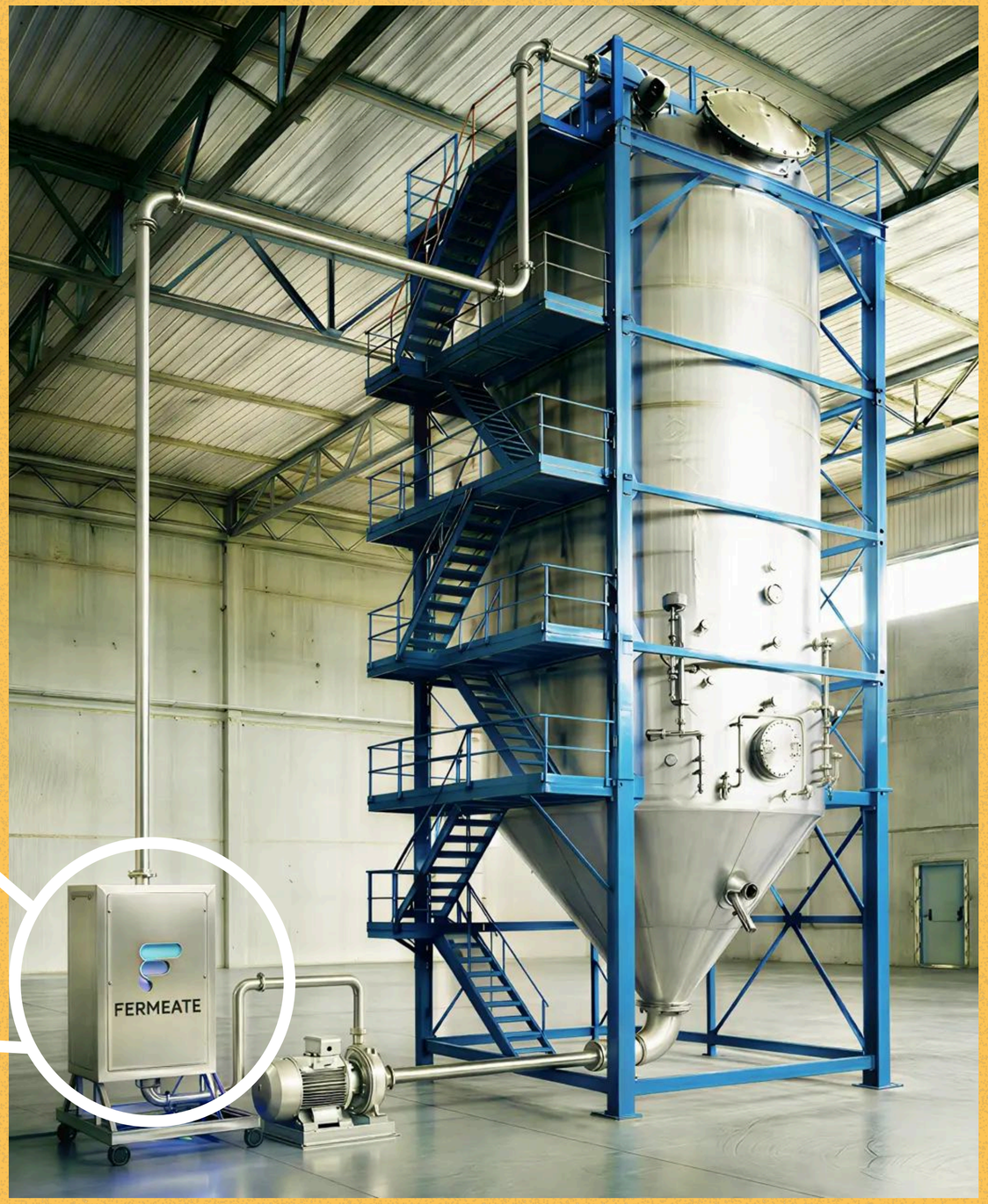
Ke Xu
Co-Founder and CEO,
PhD from Princeton

Product



identifies optimal light pattens





The Unlock

300%

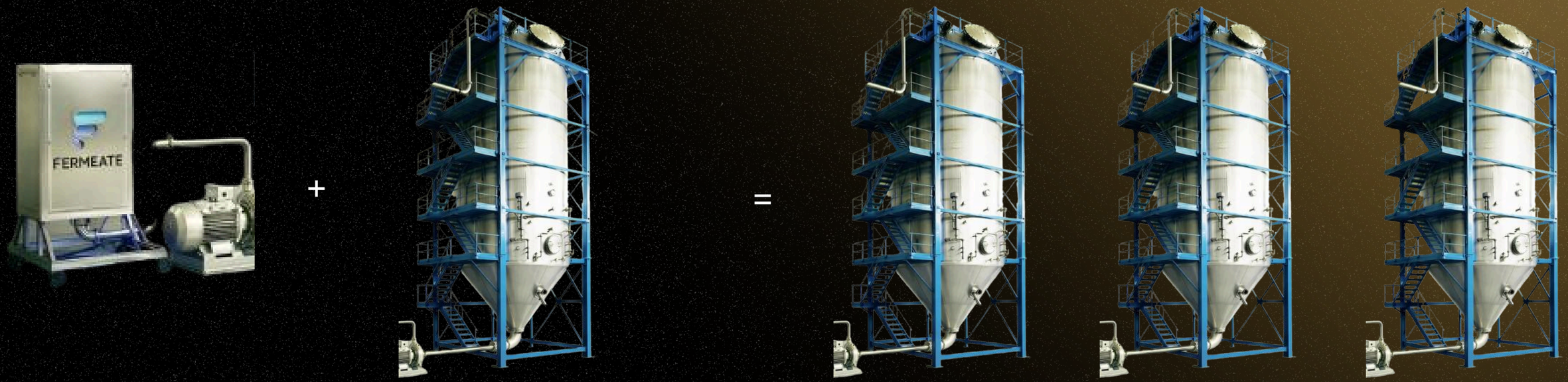
output achieved

< 5%

CapEx cost of a new bioreactor

Market

$50.75B TAM



Competitors

Fermeate is the only company that can actively manage in the middle of the fermentation process

Business Model

Selling the hardware module + ML software to industrial fermentation operators, unlocking 3x their outputs

Recession Resistant

In downturns, only low CapEx optimizations are affordable, increasing Fermeate’s competitiveness

Risk 1 — New Entrants

Prolific Machines could pivot their optogenetics technology from mammalian cells

Mitigant

Fermeate has the seminal IP in the microbial field and the licensed technology under Princeton

Risk 2 — Protein Fitting

The used microbial strains have to be retrofitted with the light sensitive proteins

Mitigant

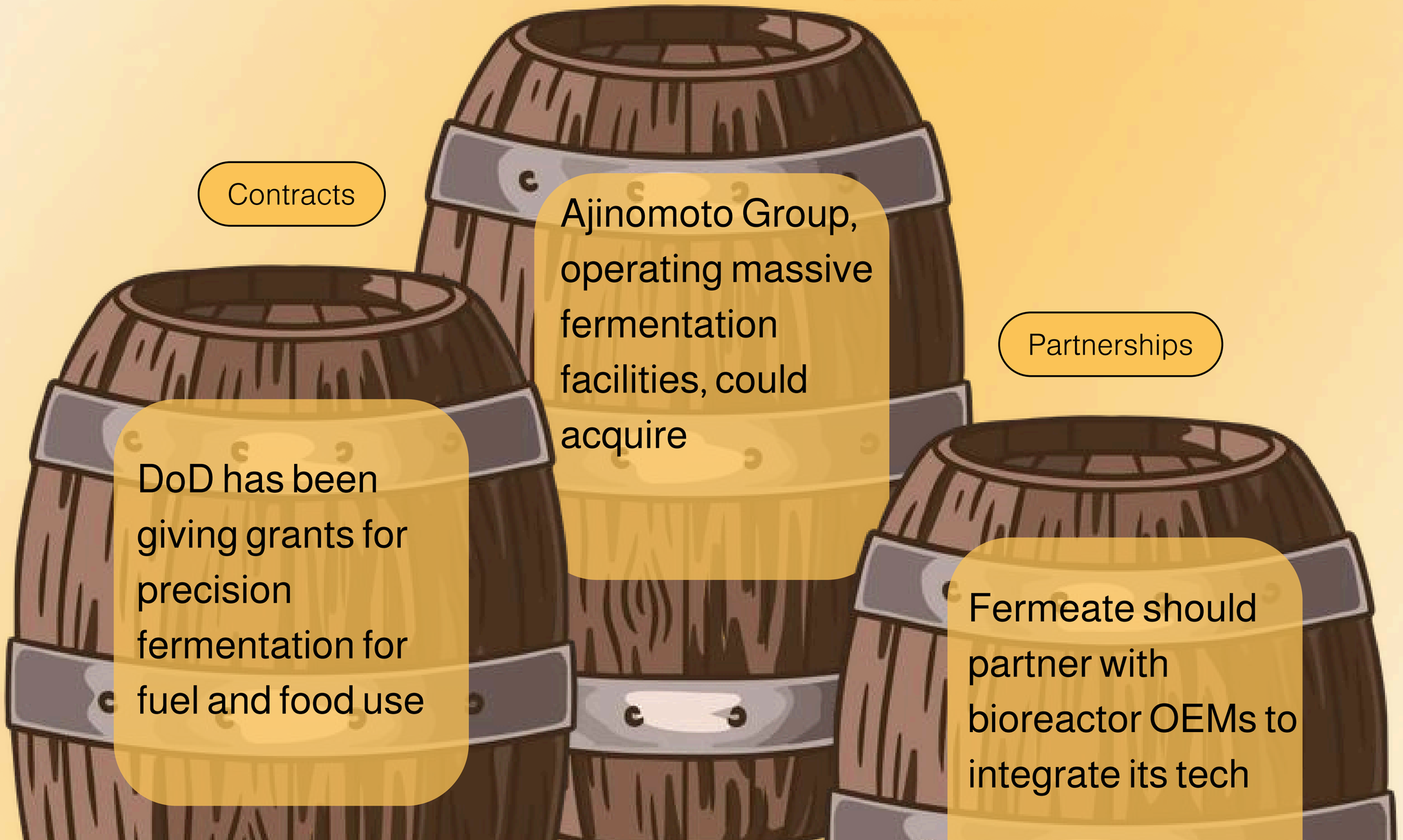
Fermeate has expanded its compatible microbial chassis to support most widely used industrial hosts

Opportunities

Exits

Contracts

Partnerships



Ajinomoto Group, operating massive fermentation facilities, could acquire

DoD has been giving grants for precision fermentation for fuel and food use

Fermeate should partner with bioreactor OEMs to integrate its tech

Thank You!

Appendix 1

< 5% of CapEx to make a new bioreactor

5% of $6M = **$300k** install cost

estimate **$115k** per year in software fees

over a **10 year** lifespan

at **35k** bioreactors globally, that's

(35k bioreactors) * ((300k install) + (115k ARR * 10 years)) =

$50.75B TAM

Appendix 2

At a Series B stage, Fermeate could realistically have **20+** deployments as they have strong traction in just Seed stage

at **$5M** revenue per facility from hardware + software,

we could see
(20 deployments) * ($5M) = **$100M in ARR**

at a moderate **10x** revenue multiple

Ajinomoto sees
($100M ARR) * (10x rev multiple) =

$1B in enterprise value